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Proposed Research Plan for UNSGC Fellowship Application

Abstract

The purpose of my research is to develop a system which can detect and track a target using passive bistatic radar and Wi-Fi signals. The tracking will be performed by detecting and tracking doppler shifts in the received signal created by the target.

Academic and Research Status

As an undergraduate student at Utah State University, I learned about continuous and discrete systems and signals, how to implement signal processing in real time on a microprocessor, and the basics of communications systems. Using what I learned, I worked with Dr. Todd Moon to develop the bistatic radar system mentioned above.

I started work for Dr. Moon May 2022 and have made significant progress towards our goals since then: We were able to receive and demodulate a transmitted OFDM signal and detect and track a doppler shift in the received signal. I believe my experience as a student at USU and the progress I have made so far on this project show I can continue to produce progress on this project.

About the Research

My research is a continuation and modification of the research by Thomas Bradshaw in his thesis titled *Alternative Doppler Extraction for Indoor Communication Signals*. Thomas found it is possible to detect doppler shifts in a received signal without knowing the transmitted signal [1]. I extended his research by applying Thomas' techniques to find doppler shifts in a multicarrier signal.

To simplify the project, a custom signal is made to mimic a Wi-Fi signal and used as the transmitted signal. The transmitted signal is made sufficiently random by generating 1 million OFDM symbols. Each OFDM symbol is made of 48 random data symbols encoded using QPSK, four known pilot symbols, and 12 zeros. The 1 million OFDM symbols are then transmitted on a loop and received by four receiver antennas.

At each receiver, the received signal is demodulated and equalized. The equalized signal is then divided into the received signal to get the channel state information for that receiver. The channel state information is then processed using a spectral estimation technique with matched filters to estimate the doppler frequency for a given time frame. The doppler frequency is then processed using a Viterbi Algorithm to remove any artifacts caused by noise. Lastly, the processed doppler frequencies from all four antennas are passed through a Kalman Filter to estimate position and velocity.

Expected Outcomes

At the end of this project, I hope to demonstrate the possibility of estimating the position and velocity of an object using a received Wi-Fi signal. I believe applying the techniques used in this project can make target tracking possible for a system at any radio frequency. For example, this technology can be used on a bigger scale to perform geolocation using signals of opportunity.

1. Bradshaw, Thomas L., "Alternative Doppler Extraction for Indoor Communication Signals" (2021). *All Graduate Theses and Dissertations*. 8111. https://digitalcommons.usu.edu/etd/8111